Drishti: Driver Drowsiness Monitoring Using Temporal Fusion of Eye Closure and Yawning

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Abstract—This paper presents the design and software verification of a temporal decision engine for camera-based driver drowsiness The engine takes into account the eye and mouth aspect ratios and combines median filtering, hysteresis, sustained eye-closure duration, There was recent yawning, and a rolling sample ratio with the eyes closed. A bounded evidence score and a prioritized decision policy produce four The states for monitoring are alert, fatigued, drowsy, and critical. The implementation uses explicit thresholds and event histories rather than a trained temporal classifier, allowing each decision to be traced based on observable cues. There are eight deterministic scenarios that are synthetic feature streams and a controlled clock verify selected state The handling of transitions, the counting of events, and memory expiry. All eight scenarios pass their specified assertions; these results establish functional The detection accuracy is not provided at the level of individual participants, based on a public dataset. a benchmark, or end-to-end latency is stated. The analysis identifies limitations associated with fixed thresholds, sampling-dependent smoothing, missing observation handling, and the difference between sample-based eye closure and physiological eyelid-closure A protocol for independent evaluation of the measurements. implementation diagrams provide a basis for subsequent experimental validation.

Index Terms—Driver drowsiness, eye aspect ratio, yawning, temporal fusion, driver monitoring.

I . INTRODUCTION

Driver monitoring requires reasoning about behavior over Time; in one image a closed eye may signify an ordinary blink, whereas a sustained closure can justify an urgent Warning: there are also many possible explanations for the mouth opening, including speaking and yawning. An useful monitoring system must therefore retain event duration and recent history rather rather than considering each image as a separate decision.

Ocular measurements offer one established basis for The Federal Highway Administration is monitoring alertness.

The researchers define PERCLOS as the ratio of the amount of time the eyes stay closed by at least 80%, noting that it is mainly concerned with slow eye closures and disqualifies mere short blinks. There is another practical element in the methods based on some landmarks of the face, whereby the authors classify eye blinks into distinct classes according to the ratios of their aspects.

YawDD offers video materials recorded while yawning and performing other activities in the car, which help to assess the second These materials are of use as behavioral indicators and stimulate the researchers in conducting their temporal analysis of the eyes and mouth, although the authors do not check the demand for this approach. This paper analyzes the Drishti time engine and analyzes whether it is possible to build a compact and transparent policy on the basis of the data it collects.

The contribution is a reproducible description of the implemented rules, a controlled functional verification, and an evaluation design that separates component behavior from the broader Assertion regarding the detection of driver drowsiness. The present contribution is an engineering prototype; its physiological validity and Comparative effectiveness is still an open question in research.

A. The problem of monitoring driver drowsiness

Drowsiness can reduce attention, slow responses to sudden Events on the road can impair a driver's judgment before they fall asleep. Sleep restriction, long shifts, and monotonous travel can Contribute to the problem; the NHTSA identifies the period from midnight to 6 A.M. and in the late afternoon are typical times for drowsy driving. There are [4] crashes. The camera surveillance looks for observable changes. early enough to support an actionable warning, while recognizing that facial behavior is an indirect indicator of alertness.

Figure 1 shows drowsiness while at the wheel, and Figure 2 does so as well. An example of a resting position in a car. The CDC identifies repeated yawning or blinking, lane drifting, and difficulty [5] The signs from recent travel should be remembered. Closing the eyes and yawning motivate the two input cues used here; head posture and vehicle movement are not measured by this The photographs and illustrations illustrate the issue. description and are not experimental samples or evidence that A person depicted was driving while under the influence.



Figure 1 is an illustration generated by AI, showing a driver resting in a parked car. This image shows the monitoring context and is not experimental data.



Figure 2. Example of a resting position at a steering wheel. Photo taken by Margo Evardson, Pexels [6], used under the Pexels License; no drowsiness diagnosis or the movement of the vehicle is deduced.

II . RELATED WORK

Landmark-based blink detection [2] uses eye geometry and Classification in terms of time. The multigranularity convolutional and recurrent framework proposed by Lyu et al. includes learned alternatives. and the blink-sequence model going with that realistic dataset of Ghoddoosian et al. [8]. Dewi et al. investigate adjusted eye-aspect-ratio thresholds [9], supporting The necessity of looking at the differences between users. More recently, YawDD+ provides finer temporal annotations for yawning The models [10]; Table I gives a summary of these sources and their implications for evaluation.

The implementation considered here uses no learned temporal The rules do explicitly differentiate between current severe closure and weights. Based on historical evidence, the decision path accessible, but creates a calibration burden: the thresholds, evidence weights, and semantic meanings of the output states must be tested; a rule-based implementation cannot be assumed to outperform learned approaches, and published results under different datasets and protocols are not a direct ranking.

III . SYSTEM DESIGN AND IMPLEMENTATION

A. Component Boundary

The implemented component is TemporalEngine in The temporal_engine.py file has an update(raw_ear, raw_mar) interface. consumes two scalar features and returns eye state, filtered features, closure duration, cumulative blink and yawn counts, recent event counts, a closed-eye ratio, target score, displayed score and status; the Python time and collections utilities handle Timing and event history; for median filtering NumPy is available. The prototype screenshots supplied with this paper show a camera view with facial landmarks, a warning-state panel and An activity log; the deterministic verification presented here evaluates the temporal engine separately; it does not measure The accuracy or the latency of the entire prototype.

Figure 3 shows the position of the temporal engine in the camera-warning processing chain. The only component that has been verified is the temporal engine. as shown by the deterministic checks mentioned here. Figures 4 and 5 Record the interface prototype provided by the authors. A reproducible evaluation must still specify landmark indices, coordinate normalization, confidence checks, camera resolution and effective sampling rate, because these choices influence the input ratios.

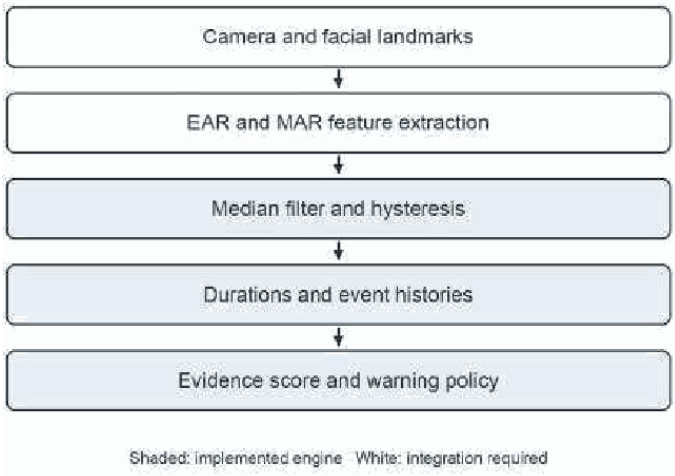


Figure 3. The processing pipeline used for evaluating the component. The shaded blocks Are owned by TemporalEngine. The camera in the surrounding area, feature-extraction and warning-display stages are outside the deterministic verification; Examples of the prototype interface can be seen in Figures 4 and 5.

The prototype landing page provides access to the monitoring The dashboard (see Fig. 4) shows the facial landmarks in the camera view. together with the present state and the numerical figures (Fig. 5). These screenshots document the implemented interface; the The values given are the output of the session shown.

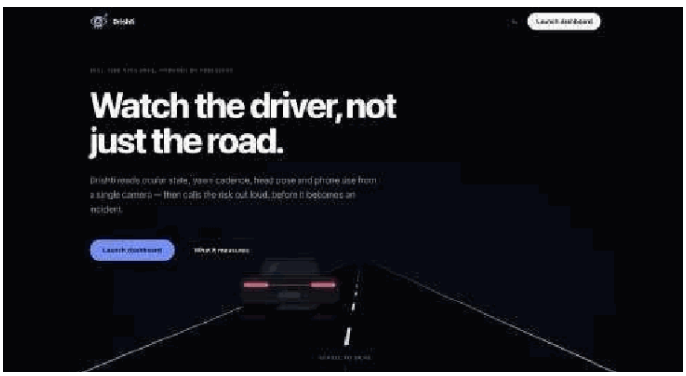


Figure 4. The landing page of the Drishti prototype together with the control used to launch the Monitoring dashboard. Screenshot provided by the authors.

Table I. Comparison of relevant research

Reference no.	Dataset used	Methodology / algorithm used	Performance parameters	Research gap / disadvantages
[1]	Controlled ocular measurement and psychomotor-vigilanc e study	Physiological PERCLOS validation	Relationship with alertness and vigilance; no project accuracy	The project needs calibrated eyelidclosure measurement before claiming equivalence.
[2]	ZJU and Eyeblink8 video sequences	Facial landmarks, eye aspect ratio, short temporal window and support vector machine	Blink detection; not a four-state drowsiness benchmark	Blink recognition alone does not Set up the fatigue or warning as valid.
[3]	YawDD in-car video sets	Dataset of normal mouth activity, speech or singing, and yawning	322 videos in one camera setup; 29 in the other; dataset descriptors	Yawning labels do not by themselves establish physiological drowsiness.
[7]	NTHU-DDD; additional FI-DDD dataset	Multi-granularity convolutional features and long short-term memory	The authors state that the accuracy is 90.05%. approximately 37 frames/s on NTHU-DDD evaluation set	Hardware and protocol differ from this project; model and inference cost require measurement.
[8]	UTA-RLDD: 60 subjects, approximately 30 h	Blink features with hierarchical multiscale long short-term memory	Three video states: alert, low vigilant, drowsy; classification benchmark	Its three labels do not directly validate The engine has four warning conditions.
[9]	Talking Face, Eyeblink8, Eye Video S1	Modified eye aspect ratio for blink detection	Blink-detection evaluation across datasets	User-dependent eye geometry motivates threshold calibration; blink The labels are still different from drowsiness.
[10]	YawDD+ frame annotations derived from YawDD	Human-verified temporal labeling; MNasNet and YOLOv11 models	v3 authors report 99.34% classification accuracy and 95.69% detection mAP	Yawn classification and objectdetection metrics cannot be transferred to drowsiness classification.

The figures mentioned above are those of the authors cited and obtained in their experimental arrangements; they are not measurements made by Drishti. The amounts in the dataset refer to scale not to predictive performance; the 2026 revision of [10] is mentioned as an arXiv preprint. So that its reported values can be traced.



Figure 5 shows a prototype camera view with a facial-landmark overlay and a The critical state is displayed. The risk panel indicates drowsiness and evidence values of 100/100; these are interface scores, not measured Accuracy in detection. Screenshot provided by the authors.

B. The definition of features and the process of filtering

For six eye landmarks p1 through p6 ordered around the eyelids, the conventional eye aspect ratio is [2]

$$EAR = (\|p2 - p6\| + \|p3 - p5\|) \text{ divided by } (2\|p1 - p4\|). \quad (1)$$

The module receives EAR rather than computing (1); it does not state whether a caller uses both eyes. It does likewise receives a mouth aspect ratio, MAR, without defining its A landmark in geometry; a front end which is to be reproduced must make this explicit. define MAR because different vertical-to-horizontal mouth The ratios have different numerical scales. This has been implemented. mouth thresholds therefore apply to the expected input These come from convention and are not universal geometric constants.

Each feature is filtered by the median of at most seven recent Observations: When initializing, the median uses the Samples are available. At a steady rate of 20 updates per second, seven The observations range over a period of 0.30 s between the earliest and the latest. A typical step usually requires four new ones. The observations should take up the entire window, which is of odd length. The filter suppresses isolated perturbations but can delay measured event onset and offset; its temporal effect changes with update rate.

C. The state of the eye and events involving closure

The eye state starts off as OPEN and turns to CLOSED when The filtered EAR is at most 0.23 and goes back to OPEN when EAR reaches at least 0.27; the values between these thresholds Keep the previous state; this hysteresis prevents repetition. The transitions occur at a single decision threshold. The code comments mention approximate webcam observations, but provide no participant count, calibration record, or independently verifiable experiment.

A closure timer begins at the very first update that is classified as CLOSED. When the eye opens again, the closure lasts from 0.08 through The cumulative blink count increases in 0.65 second increments. A closure When it lasts for at least 0.8 seconds it produces one long-closure event It goes on; the event is not recorded more than once during that period. The closure period ranges from 0.65 to 0.8 s and lies outside both Definitions of events. All the timings refer to the filtered state, which can vary from the basic physical movement of the eyelid.

Figure 6 summarizes the two transitions between eye states and the A band that is intermediate and retains the previous state.

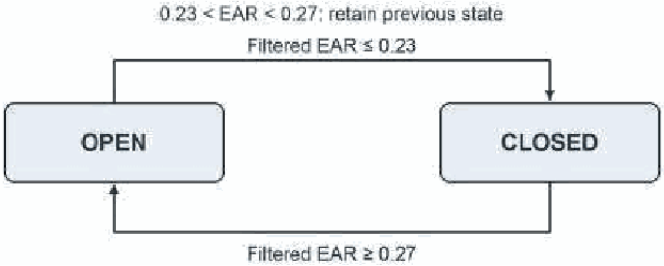


Figure 6 shows the eye-state hysteresis using the implemented filtered EAR. thresholds.

D. State and Event Memory – Yawn

The yawn candidate begins when the filtered MAR reaches at least 0.42. It It is only confirmed when that condition continues for at least 0.65 seconds. Confirmation increments the cumulative yawn count and Records the timestamp of an event. Once it has been confirmed, the active yawn stops when MAR drops to 0.28 or below. Therefore, the lower stopping threshold preserves an active event through Intermediate mouth openings; a brief crossing of the threshold that If it does not meet the duration requirement, it does not produce a yawn event.

Yawn events remain in a 45 s history, while long closures Stay within a 30-year history; if events are too old then they are removed. exceeds the corresponding window; an event exactly on the The boundary will stay in place until the following qualifying update. Counts therefore represent recent event onsets or registrations rather longer than the duration of the event. A very long active yawn can outlive its event-memory entry.

E. Proxy for Rolling Eye Closure

At each update, the engine stores a timestamp and binary eye Then the state keeps the entries from the previous 30 seconds. Its quantity named perclos is

Psample(t) = (Σ ci) / N, where ci is either 0 or 1, and ti is at least t minus 30 seconds. (2) ∈

Here, ci denotes the thresholded closed state and N is the The number of updates kept. Equation (2) is an example It offers an approximation of a temporal proportion only when The observations are at regular intervals and involve short blinks. uses the available history during startup, and does not measure With 80 per cent eyelid coverage, it ought to be described as a PERCLOS-inspired proxy rather than an established A physiological PERCLOS measurement [1]. The initial one was closed sample immediately yields a ratio of 1, despite the absence of a complete observation window.

IV. EVIDENCE FUSION AND WARNING POLICY

A. Bounded Evidence Score

Let d be current closure duration, P the sample ratio, Y recent Yawns, due to the recent long closures, and I have the active-yawn indicator. The target score is

T is equal to the minimum of 100 and the sum of C(d), E(P), W(Y), A(I), H(L) and B. (3)

Closure evidence C contributes 20, 45, 65, or 85 points when It reaches 0.8, 1.5, 2.0, or 3.0 seconds respectively by using only the The interval with the highest level of satisfaction is Eye-ratio evidence E, which amounts to 10, The score is 18, 25, or 30 when the P values are 0.20, 0.30, 0.40, or 0.50. W contributes 10 for one recent yawn, 20 for two, and 30 for Three at least; an active yawn gives five points. H contributes eight points for one long closure, 18 for two or three, and 25 For a minimum of four; all the rest are zero.

A ten-point bonus B is applied in the case where d is 0.8 s or longer and Either Y is two or more and P is at least 0.35. At first, the implementation checks the yawn condition and afterward it checks for the ratio condition, thus ensuring that it never grants both bonuses at once. Weights in this case are given through manual engineering choices. The final score, which can go anywhere from 0 to 100, is evidence score and not an estimate of likelihood or percentage of yawning.

B. The score and recovery shown

The displayed score S follows an asymmetric recurrence:

S* = Sprev plus α times (T minus Sprev). (4)

When T goes above the previous score the coefficient α is 0.20. and 0.08 in all other cases. If the eyes are open and there is no active yawn, On that update an extra 0.12 is deducted. The value is clipped to [0, 100] and rounded to an integer for the reported The coefficients apply at each update rather than at each The second point is that the higher the update rate, the faster the score becomes. Evolution in terms of physical time; a later revision may use elapsed time coefficients once equivalent behaviour has been verified.

B. Prioritized Status Selection

The decision policy first assigns CRITICAL if d is 3.0 seconds or more. For d greater than or equal to 2.0 seconds together with at least two recent yawns. If DROWSY is chosen when d is greater than or equal to 1.5 seconds; Alternatively, P is greater than or equal to 0.45 with at least two recent long closures. Also counts as a valid case; the third DROWSY rule stipulates that the score must be at least 55. along with P being greater than or equal to 0.35 or d being greater than or equal to 0.8 s. These conditions are assessed before the fatigue rules.

If no higher state applies, FATIGUED is assigned when at Either at least two yawns are still remembered or the score gets to 30. In all other cases the state is ALERT. The names in question are operational. They do not assign output labels, and the four-state policy does not directly match the three labels in UTA-RLDD [8]: A current severe closure can trigger CRITICAL before the smoothed score reaches a high value, while historical evidence can keep lower-level warnings active after the eyes reopen.

The timeline of activity (Fig. 7) on the dashboard shows information about fatigue, drowsiness, critical state, and yawning along with relevant time stamps. The displayed counters provide a summary of the demonstration session. The label on the interface saying "driver unresponsive" reflects the output of the system while independent observations are required to reveal the real condition of the driver.

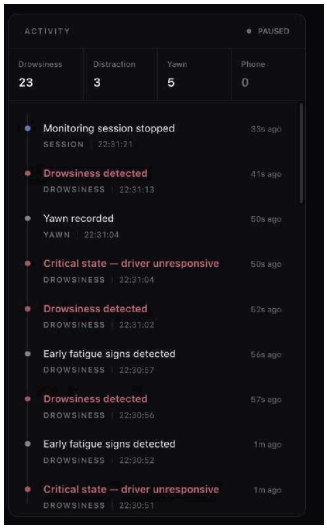


Fig 7. Prototype activity panel after monitoring has been stopped. The screenshot shows the warnings and yawn occurrences with time logs as well as numbers indicating 23 instances of drowsiness, 3 instances of distraction, 5 instances of yawning, and 0 instances of phone activity. It is important to note that these counts are not a measure of accuracy. The authors provided the screenshot.

IV. FUNCTIONAL VERIFICATION

A. Deterministic Test Method

The engine has been tested unchanged using synthetic EAR and MAR at 20 updates/second. To prevent any dependence on execution speed, the patched clock was used so that the timestamps began from a known reference. For each run, a new engine was utilized. For Openeye, the input value was 0.35; for closed-eye it was 0.15; for neutral mouth, it was 0.15; and for open mouth it was 0.60. Both the verification script and machine-readable results have been kept with this paper. No images from cameras, volunteers or labels from the publicly available dataset were utilized in the tests.

A total of eight scenarios were put through testing: (1) the eye closed eye state; (2) hysteresis, which consists of closed, intermediate, and opening EAR values; (3) a short blink; (4) transitioning from a sustained closure to DROWSY and CRITICAL; (5) rejecting a quick mouth opening; (6) two yawns resulting in a FATIGUED status; (7) a yawn leaving consciousness while it still counts; and (8) a closed eye cold start. All of the scenarios were successful in their explicit claims.

Figure 8 illustrates another synthetic trace from the standard engine at 20 updates/s. The raw EAR can take values from 1.05 to 5 s as 0.15 and equals to 0.35 otherwise, while MAR is set to 0.15. Application of median filtering leads to shifts in the output values of closure opening as well as reopening.

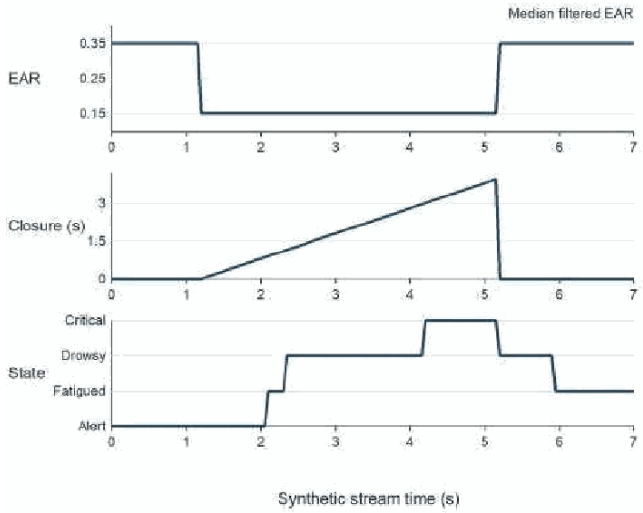


Fig. 8. The engine reacts when it comes to a controlled synthetic closing and reopening. The trajectory represents how the software behaves, not the outcome of participant or dataset detection.

B. Interpretation of Verified Behavior

The baseline managed to maintain ALERT without either score or closed-eye ratio. The hysteresis performed CLOSED, generating EAR value equal to 0.25 and reopening it at the value of 0.30. The brief closure activity led to the partial count of blinks without establishing the presence of the closure. The closure event continued for some time producing DROWSY, and then CRITICAL statuses, although the length count remained equal to one. The long-lasting mouth opening event resulted in the generation of separate events only after the mouth came back to the initial state.

The cold-start event produced a value for P equal to 1 and closure length equal to zero but produced ALERT. This means that the high startup ratio alone does not determine the status since the ratio only refers to one single measurement. The experiments represented in this table illustrate particular characteristics of the state-machine algorithm. Eight shown scenarios cannot indicate the effectiveness of detection although they pass successfully.

VI. PROPOSED DATASET EVALUATION

A. Resources and Labeling

Figures 4, 5 and 7 document the current prototype interface: its landing page, camera-and-state display and session activity timeline. Figures 1 and 2 illustrate the road-safety problem only. A future evaluation should also document camera placement, recording conditions, actual dataset provenance and independently assigned sample labels. The implementation screenshots do not replace that evaluation record.

UTA-RLDD can be used as a reference in analyzing drowsiness behavior, whilst YawDD and YawDD+ can be applied for yawning analysis. However, these tools have varying functions and labels. It is crucial that yawning detection should not be grouped with the detection of the drowsiness state. The final report describing the datasets collected in the course of this project must handle recruitment, recording conditions, camera position, glasses, lighting, frame rate, evaluation of the reference labels, and any specificities related to labeling four types of alerts.

The training, calibration, and test datasets must be taken from different subjects. A calibration dataset is necessary, even for a rule-based system. Frames or videos taken from one person must not belong to both calibration and test datasets. As for threshold and score decisions, all these must be made prior to the conclusion of the final test evaluation, as it is vital to state which datasets have been used for the experiment in the paper, thus avoiding the practice of presenting the literature list as experimental proof.

B. Metrics and Comparisons

For making decisions involving binary events, one can measure precision as TP/(TP+FP), recall as TP/(TP+FN), and the F1 score by the equation 2PR/(P+R), where P indicates precision and R indicates recall. For processing multiple classes, one has to measure the confusion matrix, calculate macro-average F1 scores, recall for each of the individual classes, and calculate balanced accuracy. When matching events, one has to define an acceptable amount of time for onset and avoid considering every frame of a single yawn as a separate event. In addition to this, the confidence interval should be expressed in a way that reflects variation between different subjects rather than suggesting that highly correlated frames can be seen as independent values.

Ablations ought to make comparisons using the instantaneous eye thresholds only, the eye-duration rules, the eye and mouth policy together with the use of closure ratio, and the complete eye-mouth policy. Separate ablations may be conducted by taking out median filtering, hysteresis, and asymmetric score filter from all approaches. The comparison must employ the same subjects, features, sampling assumptions, and metrics. Both successes and failures should be reported instead of simply picking only the good clips.

VII. DISCUSSION AND LIMITATIONS

The engine provides a relatable temporal policy with outputs that can correlate with definite durations and events. The functional outcome backs the assertion that the chosen rules work as described. It does not indicate that the threshold enables to differentiate people regarding physiological fatigue. However, eye geometries, facial motion, camera angle, glasses, lights, and mouth ratio use may interfere with the information supplied. Individual adjustments must be checked with the fixed threshold first before any application.

Several implementation issues influence reproducibility. Timestamps are taken from the wall clock rather than a monotonically running time or the recorded time of the video. In this situation, if a video gets played back at a higher speed than real-time, the results would also be different if no measures are taken to avoid that situation. Problems caused by the absence of required characteristics are not clearly stated. If there was a closed state indicated when closing a gap, then the time needed to close the gap again using the previous state would increase. An integration layer would need to ensure the validity of the observations, handling the gaps, and indicating when the data is unavailable.

The closed-eye proxy looks at the samples equally while different samples comprise unique histories. The history of samples is composed during the 30 s interval showing about 30f entries with close frequency. This means that the processing time for the closed flags becomes the same as the number of entries kept in the history while the events producing changes in the histories also require time for computation.

Behavioral indicators too may comprise ambiguities. Speaking may be mistaken for yawning, and closing one's eyes voluntarily does not indicate drowsiness. Cutoff points with a fixed length can fail to account for unusual blinking or categorize ongoing behavioral action incorrectly. These restrictions give rise to the requirements for subject disjoint validation, clear reporting of failures, and carrying out a comparison with independent reference labels. The prototype image and composite images should represent the settings employed for the conduct of such evaluations.

VIII. CONCLUSION

Drishti utilizes temporal fusion of the eye and mouth-based signals through median filtering, hysteresis, event histories, a bounded evidence score, as well as prioritized warning rules. The chosen behaviors were validated in eight controlled synthetic experiments of single-event counting, progression through sustained closure, and other scenarios. At the present time, the evidence indicates that the function of the temporal decision engine has been confirmed although the discrimination at the dataset level, false warnings, and end-to-end timing have yet to be assessed. The subsequent stage of experiments must include the integration of the landmark front end, recording of the actual datasets provenance, validation through independent label assessment, and testing of the full prototype under a protocol independent of subjects.

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